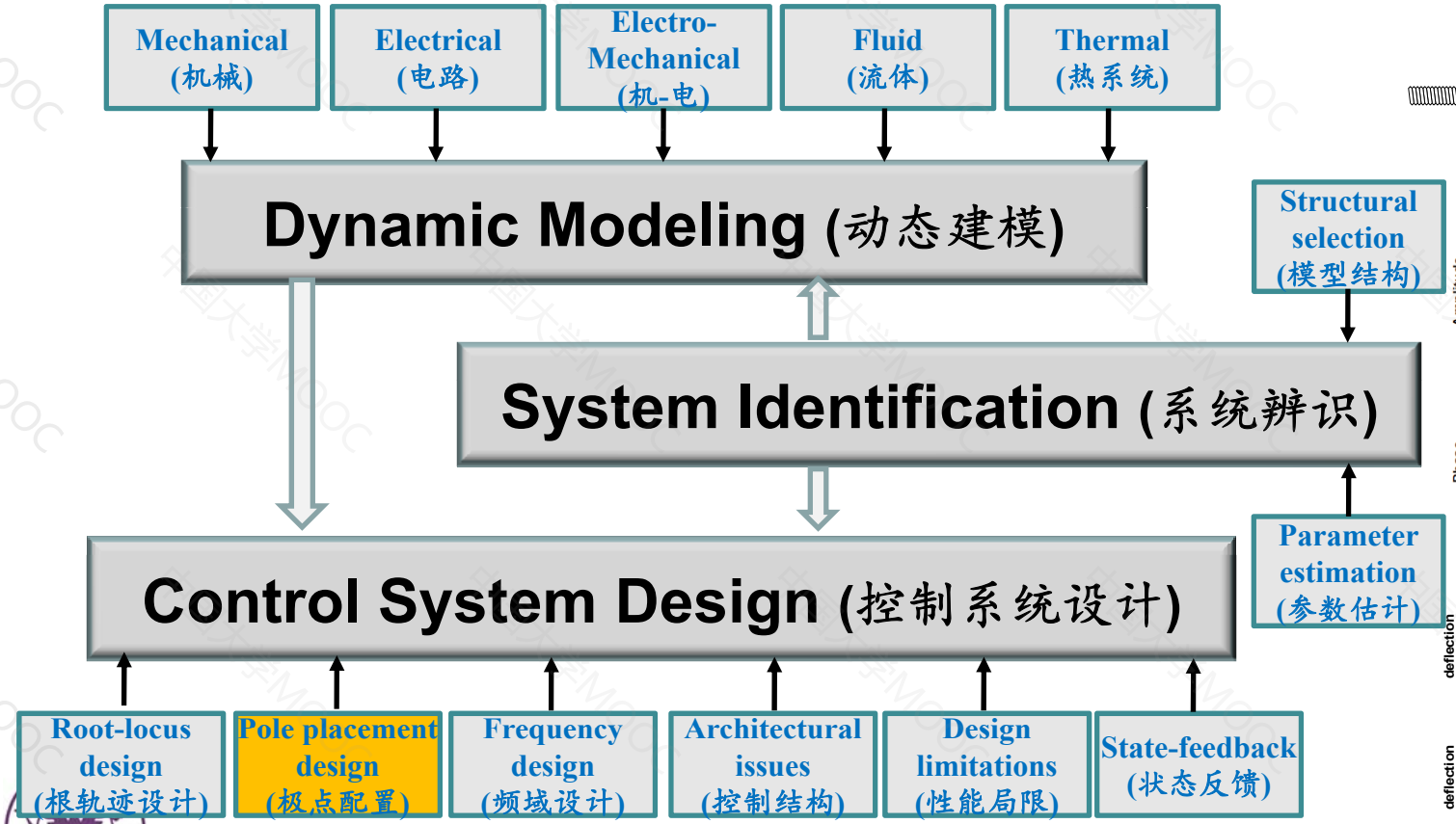
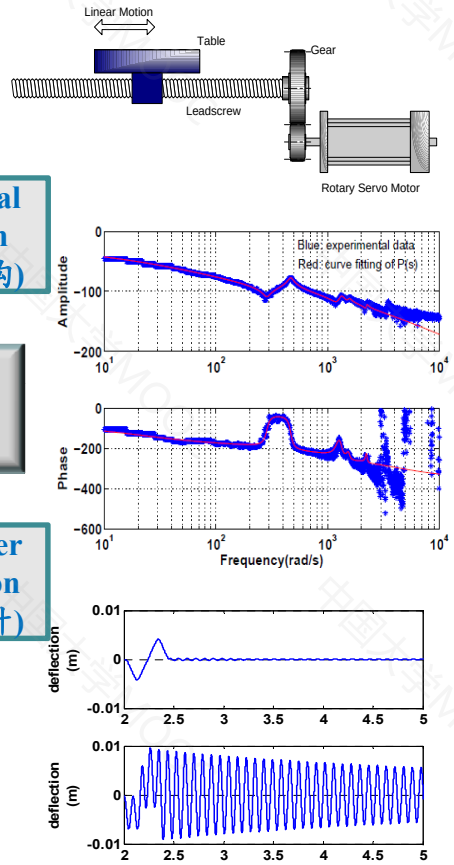
Controller Design via Pole Placement – Principle

(通过极点配置的控制器设计 – 原理)

Main Contents of Course (课程主要内容)





Pole Placement

- **Nominal Closed Loop Characteristic Equation** (名义闭环特征方程)

where $D_{pn}(s)D_c(s)+N_{pn}(s)N_c(s)=0$

$$N_{pn}(s)=b_{pm}s^m+b_{p(m-1)}s^{m-1}+\cdots+b_{p1}s+b_{p0}$$ (assume $m<n$)

$$D_{pn}(s)=s^n+a_{p(n-1)}s^{n-1}+\cdots+a_{p1}s+a_{p0}$$

$$N_c(s)=b_{cm_c}s^{m_c}+b_{c(m_c-1)}s^{m_c-1}+\cdots+b_{c1}s+b_{c0}$$ $m_c<n_c$

$$D_c(s)=s^{n_c}+a_{c(n_c-1)}s^{n_c-1}+\cdots+a_{c1}s+a_{c0}$$

m_c+1 controller coefficients for N_c: $b_{cmc},\ldots,b_{c0}$
(N_c有m_c+1个控制器参数)

n_c controller parameters to choose for D_c: $a_{c(nc-1)},\ldots,a_{c0}$
(D_c有n_c个控制器系数)

- **Nominal Closed Loop Characteristic Polynomial** (名义闭环特征多项式)

$$A_C(s)=D_{pn}(s)D_c(s)+N_{pn}(s)N_c(s)$$
$$=s^{n_L}+a_{Cn_L-1}s^{n_L-1}+\cdots+a_{C1}s+a_{C0}$$

Order $m+m_c<n+n_c=n_L$

Pole Placement

- **Desired Closed Loop Characteristic Polynomial**

(期望闭环特征多项式)

$$A_{Cd}(s) = (s - p_{1d})(s - p_{2d})\cdots(s - p_{n_L d})$$
$$= s^{n_L} + a_{C(n_L-1)d}\, s^{n_L-1} + \cdots + a_{C1d}\, s + a_{C0d}$$

- **General Formulation of Pole Placement**(极点配置的一般公式)
  - Given $P_n(s)$ and any desired CL pole locations, i.e., known n_L and $A_{Cd}(s)$ but with coefficients being arbitrarily specified, does there exist a proper $C(s)$ that makes $A_C(s)$ being the desired $A_{Cd}(s)$?

(给定$P_n(s)$和任意期望闭环极点位置，即已知n_L和可任意指定系数的$A_{Cd}(s)$，是否存在一个合适的$C(s)$，使$A_C(s)$成为期望的$A_{Cd}(s)$)

Sylvester's Theorem

- **Coprime Condition for Two Polynomials** (二个多项式的互质条件)

Given two polynomials
(给定两个多项式)

$$N(s) = b_n s^n + b_{n-1} s^{n-1} + \cdots + b_1 s + b_0$$
$$D(s) = a_n s^n + a_{n-1} s^{n-1} + \cdots + a_1 s + a_0$$

Form $2n \times 2n$ Sylvester matrix
(形成2n×2n的西尔维斯特矩阵)

Then

$$S = \begin{bmatrix} \left.\begin{matrix} a_n & 0 & \cdots & 0 \\ a_{n-1} & a_n & \ddots & \vdots \\ a_{n-2} & a_{n-1} & \ddots & 0 \\ \vdots & a_{n-2} & \ddots & a_n \\ a_0 & \vdots & \ddots & a_{n-1} \end{matrix}\right\} n+1 & \left.\begin{matrix} b_n & 0 & \cdots & 0 \\ b_{n-1} & b_n & \ddots & \vdots \\ b_{n-2} & b_{n-1} & \ddots & 0 \\ \vdots & b_{n-2} & \ddots & b_n \\ b_0 & \vdots & \ddots & b_{n-1} \end{matrix}\right\} \\ \left.\begin{matrix} 0 & a_0 & & a_{n-2} \\ \vdots & \ddots & \ddots & \vdots \\ 0 & \cdots & 0 & a_0 \end{matrix}\right\} n-1 & \left.\begin{matrix} 0 & b_0 & & b_{n-2} \\ \vdots & \ddots & \ddots & \vdots \\ 0 & \cdots & 0 & b_0 \end{matrix}\right\} \end{bmatrix}$$

n $\quad$ n

***N*(s) and *D*(s) are coprime iff det(S) ≠0 !** (当且仅当 det(S) ≠0，*N*(s)和*D*(s)互质)

SISO Polynomial Approach

- **Lemma** (引理)

Assume that $N_{pn}(s)$ and $D_{pn}(s)$ are coprime (no common factor). Then, as long as order of desired CL polynomial $A_{Cd}(s)$ is no less than $2n$-1, there always exists a proper controller $C(s)$ that solves the pole placement problem (假设$N_{pn}(s)$和$D_{pn}(s)$是互质的(没有共同的因子)，只要期望闭环特征多项式$A_{Cd}(s)$ 的阶数不小于$2n$-1，那么总存在一个适当的控制器$C(s)$来解决极点配置问题)：

$$D_{pn}(s)D_c(s) + N_{pn}(s)N_c(s) = A_{Cd}(s) \qquad \text{(PA1)}$$

In fact, when $n_L = 2n$-1, the solution is unique with coefficients of $C(s)$ of order n-1 solved by $n_c = n$-1 (事实上，当$n_L = 2n$-1时，通过使$n_c = n$-1解得的n-1阶$C(s)$系数是唯一的)

$$2n-1\left\{\begin{bmatrix} 1 \\ a_{c(n-2)} \\ \vdots \\ a_{c0} \\ b_{c(n-1)} \\ \vdots \\ b_{c0} \end{bmatrix}\right. = S^{-1}\begin{bmatrix} 1 \\ a_{C(2n-2)d} \\ \vdots \\ a_{Cnd} \\ a_{C(n-1)d} \\ \vdots \\ a_{C0d} \end{bmatrix} \quad \text{with} \quad S = \begin{bmatrix} 1 & & & & 0 & & & \\ a_{p(n-1)} & \ddots & & & b_{p(n-1)} & \ddots & & \\ \vdots & & \ddots & & \vdots & & \ddots & \\ a_0 & & & 1 & b_0 & & & 0 \\ & \ddots & & a_{p(n-1)} & & \ddots & & b_{p(n-1)} \\ & & \ddots & \vdots & & & \ddots & \vdots \\ & & & a_0 & & & & b_0 \end{bmatrix}$$

SISO Polynomial Approach

Case 1: $n_L = 2n - 1 + k$

→ The controller order is (控制器阶次) $n_c = n-1+k, k>0$

Then, we have n_c controller parameters to choose for $D_c(s)$ and m_c+1 parameters for $N_c(s)$
(那么，我们可以为$D_c(s)$选择n_c个控制器参数和为$N_c(s)$选择m_c+1个参数)

→ A total of $n_c + m_c + 1 = n - 1 + k + m_c + 1$ coefficients free to choose (总共可自由选择$n_c + m_c + 1 = n - 1 + k + m_c + 1$个系数)

→(i) $n_c + m_c + 1 \le 2n_c+1 = 2(n - 1 + k)+1 = 2n - 1 + 2k = n_L + k$, (since $m_c \le n_c$)

→ (ii) But to match $A_c(s)$ with the desired $A_{cd}(s)$, we only have to make sure $a_{c(nL-1)},\ldots, a_{c0}$ equal to $a_{c(nK-1)},\ldots, a_{c0}$, which represent n_L equations to solve (为使$A_c(s)$和$A_{cd}(s)$相匹配，我们必须使$a_{c(nL-1)},\ldots, a_{c0}$ 等于$a_{c(nK-1)},\ldots, a_{c0}$，表明有$n_L$个方程需要求解)

→ The number of free controller parameters (自由控制器参数个数) $2n - 1+2k=n_L+k \ge$
The number of coefficients to match, which is n_L (需要匹配的系数个数n_L)

→ As long as $k \ge 0$, infinite solution, when $k=0$, the solution is unique (只要$k \ge 0$无穷解，当 $k=0$时唯一解).

Case 2: $n_L < 2n - 1$

→ Can not arbitrarily place the desired poles (不能任意配置期望极点)

Constrained Pole Placement

- **Controller with Integration** (带积分的控制器)

Want(希望) $D_c(s) = s\bar{D}_c(s)$

Pole Placement Problem (极点配置问题)

$$D_{pn}(s)s\bar{D}_c(s) + N_{pn}(s)N_c(s) = A_{Cd}(s)$$

Equivalent Pole Placement Problem (等效极点配置问题)

$$\bar{D}_{pn}(s)\bar{D}_c(s) + N_{pn}(s)N_c(s) = A_{Cd}(s) \qquad \text{(PA2)}$$

which can be solved as before by assuming an equivalent fictitious plant of order $n+1$ and a new denominator of (通过假设一个等效的假想n+1阶系统和新的分母来解决)

$$\bar{D}_{pn}(s) = sD_{pn}(s)$$

Solution always exists if n_L is no less than $2n$. When $n_L=2n$, the solution is unique with order of $\bar{D}_c(s)$ being n-1 and order of $N_c(s)$ being n ! (如果n_L不小于$2n$，结果总是存在的。当 $n_L=2n$ 时结果是唯一的：$\bar{D}_c(s)$ 的阶数是n-1， $N_c(s)$的阶数是n。)

Constrained Pole Placement

- **Controller with Stable Pole/Zero Cancellation(稳定零极点相消控制器)**

Q: Want to cancel some stable poles/zeros of plant TF
(想要消掉系统传函中一些稳定的极点和零点)

Example:

$$D_{pn}(s) = (s - p_c)\bar{D}_p(s) \qquad\qquad D_c(s) = (s - z_c)\bar{D}_c(s)$$
$$\Rightarrow$$
$$N_{pn}(s) = (s - z_c)\bar{N}_p(s) \qquad\qquad N_c(s) = (s - p_c)\bar{N}_c(s)$$

Pole Placement Problem**(极点配置问题)**

$$= (s - p_c)(s - z_c)[\bar{D}_p(s)\bar{D}_c(s) + \bar{N}_p(s)\bar{N}_c(s)]$$

$$(s - p_c)\bar{D}_p(s)(s - z_c)\bar{D}_c(s) + (s - z_c)\bar{N}_p(s)(s - p_c)\bar{N}_c(s) = A_{Cd}(s)$$

which has a solution only if $A_{Cd}(s)$ contains the cancelled poles and zeros **(只有 $A_{Cd}(s)$ 包含消掉的极点和零点时才有解):**

$$A_{Cd}(s) = (s - p_c)(s - z_c)\bar{A}_{Cd}(s)$$

Equivalent Pole Placement Problem **(等效极点配置问题)**

$$\bar{D}_p(s)\bar{D}_c(s) + \bar{N}_p(s)\bar{N}_c(s) = \bar{A}_d^c(s) \qquad \text{(PA3)}$$

Cancelled poles/zeros remain as CL poles !
(消掉的极点/零点仍然是闭环极点)

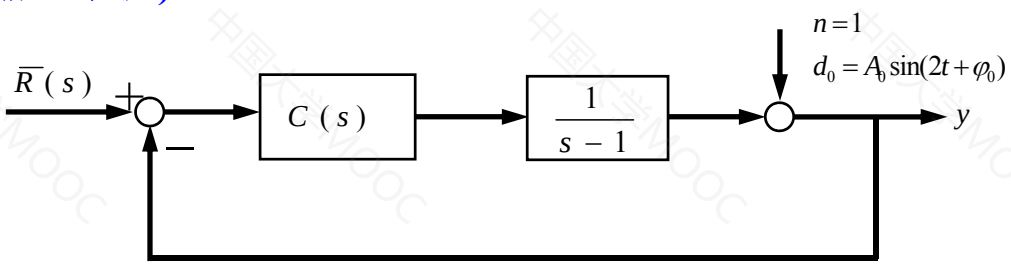
Example of Constrained Pole Placement

- **Problem Formulation** (问题描述)

Want to completely reject a sinusoidal output disturbance of 2 rad/s of unknown amplitude and phase for a nominal plant of (想要使名义系统能完全抑制一个未知振幅和相位的**2 rad/s**的正弦输出干扰)

$$P_n(s) = 1/(s-1)$$

A_0, φ_0 are unknown



- **Equivalent Design Performance Specifications** (等效设计性能指定)

$$Y(s) = S(s)d_0(s) \text{ where } S(s) = \frac{1}{1+L(s)} = \frac{1}{1+C(s)P(s)} = \frac{D_c D_p}{D_c D_p + N_c N_p}$$

$$\text{for } d_0 = A_0 \sin(2t+\varphi_0), \quad y(t) \to y_{ss}(t) = A_0 |S(j2)| \sin(2t + \varphi_0 + \angle S(j2))$$

Then, as long as $|S(j2)| = 0 \Leftrightarrow |D_c(j2)| = 0$, no s.s error caused by $d_0(t)$

Example of Constrained Pole Placement

- **Constraints on Controller to Meet the Design Requirements (约束控制器满足设计要求)**

$$D_c(s)=(s-j2)(s+j2)\bar{D}_c(s)=(s^2+2^2)\bar{D}_c(s)$$

Then as long as $n_c \geq 2$, then, we can arbitrarily place all CL poles. For example, choose n_c=2

(只要$n_c \geq 2$，那么，我们可以任意配置所有的闭环极点。例如，选择n_c=2)

$$C(s)=\frac{b_{c2}s^2+b_{c1}s+b_{c0}}{(s^2+2^2)}$$

- **Pole Placement Design (极点配置设计)**

Suppose that we would like to place all CL pole at $P_{CL1,2,3,d}$=-3 **(假设我们想要配置所有的极点在$P_{CL1,2,3,d}$=-3)**

$$A_{CLd}(s)=(s+3)^3=s^3+9s^2+27s+27$$

$$A_{CL}(s)=D_pD_c+N_pN_c=(s^2+2^2)(s-1)+1\times(b_{c2}s^2+b_{c1}s+b_{c0}) \Rightarrow \begin{cases}-1+b_{c2}=9\\4+b_{c1}=27\\-4+b_{c0}=27\end{cases} \text{ or } \begin{cases}b_{c2}=10\\b_{c1}=23\\b_{c0}=30\end{cases}$$

$$=s^3+(-1+b_{c2})s^2+(4+b_{c1})s+(-4+b_{c0})$$

then, $A_{CL}(s)=A_{CLd}(s)$, i.e., all CL poles are at -3.

Summary

- **Motivation** (动机)
  - A systematic procedure to synthesize a controller for SISO LTI systems to place poles of nominal closed-loop system at certain desired locations (为SISO线性时不变系统设计控制器的一种系统性过程以使名义闭环系统的极点配置在某些期望位置上)
- **Polynomial Approach** (多项式方法)
  - General Formulation (一般形式)
    - No constraint on the controller to be used (对控制器没有约束)
  - Practical Constraints (实际约束)
    - Controller with integrator (带积分的控制器)
    - Controller with stable pole/zero cancellation (稳定零极点相消的控制器)